

Quantum Dot Solar Cells (QDSCs): A Review of Next-Generation Photovoltaics

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Abstract—Quantum Dot Solar Cells (QDSCs) are next-generation alternatives to conventional crystalline silicon (Si) solar cells, which are capped by the Shockley-Queisser efficiency limit to energy conversion efficiencies of approximately 30%. QDSCs take advantage of the unique properties of quantum dots (QDs)—nanoscale semiconductor particles that exhibit quantum confinement effects—enabling tunable bandgaps and multiple exciton generation (MEG). This paper examines how QDSCs can absorb a wider range of the solar spectrum, generate multiple excitons per incident photon, and leverage advanced QD layering techniques to achieve theoretical energy conversion efficiencies exceeding 60%. It also reviews the critical challenges QDSCs face, including manufacturing-induced surface defects that degrade their performance to real-world conversion efficiencies closer to 16%, the use of toxic materials, and complicated manufacturing processes. Overall, the paper highlights that with further targeted research, QDSCs can feasibly overcome conventional solar cells and provide a reliable avenue for more sustainable and efficient solar energy production.

Keywords—Solar Cells, Quantum Dots, Quantum Dot Solar Cells, Tunable Bandgap, Multiple Exciton Generation, Photovoltaics, Quantum Confinement, Energy Conversion Efficiency, Solar Energy, Quantum Dot Layering

I. INTRODUCTION

A. Problem Statement

Solar cells, also known as photovoltaic cells due to their photovoltaic properties, are used in solar panels and arrays to convert sunlight into electricity [1]. Solar cells are increasingly recognized as a viable solution for producing renewable and clean energy, particularly amid the ongoing climate change crisis [2]. They can convert sunlight directly into electricity without additional processes [1]. However, current-generation solar cells—single-junction solar cells—yield relatively low energy conversion efficiencies of around 30%—meaning only 30% of the incident light energy is converted into electrical energy, while the rest is lost through different loss mechanisms [1], [3]. Single-junction solar cells are the most widely utilized cell technology [4]. Improving their conversion efficiency has been a major focus of research efforts to make solar energy production more cost-effective and commercially viable.

Over the past two decades, various next-generation solar cells with high theoretical conversion efficiencies have been proposed, including quantum dot solar cells (QDSCs) [2]. QDSCs leverage the unique properties of quantum dots (QDs)—nanometer-scale semiconductor crystals [5].

The unique properties of QDs, such as their tunable bandgap, enable QDSCs to absorb a wider spectrum of light and efficiently convert it into electricity [5]. QDSCs demonstrate the potential to exceed the Shockley-Queisser limit—the maximum theoretical efficiency of a single-junction solar cell—and achieve high theoretical conversion efficiencies of upwards of 63% [2]. They have the potential to change the landscape of solar energy production. On the other hand, QDSCs, similar to other next-generation solar cells, have achieved much lower energy conversion efficiencies in experimental studies due to manufacturing challenges, among other factors [6]. More research must be done to address their limitations.

B. Motivation

QDs and their application in QDSCs are the focus of the paper. The paper aims to understand how QDs are leveraged by QDSCs to improve existing solar cell technologies.

C. Breakdown

The paper is organized into three sections:

- Section II presents background on the need for renewable energy, the demand for solar energy production, the functioning of solar cells, and the limitations of current-generation solar cells.
- Section III covers QDs and their unique characteristics.
- Section IV focuses on QDSCs, exploring how they can overcome the limitations of current-generation solar cells, their performance compared to other next-generation solar cells, their construction, environmental considerations, and more.
- Section V discusses the future work for QDSCs.

Overall, the paper reviews the strengths, weaknesses, potential, and challenges associated with QDSCs. It offers readers an understanding of the current state of QDSC technology and future research directions.

D. Literature Review Methodology

A comprehensive literature review was conducted using databases like Scopus. Keyword searches included queries like "quantum dot" AND "solar cell," "quantum confinement" AND "quantum dot," and "quantum dot solar cells" OR "QDSCs." The review focused on a blend of seminal and recent peer-reviewed articles.

II. BACKGROUND

This section provides an overview of the basics of solar energy and solar cells to help the reader understand the need for next-generation solar cell technologies like QDSCs. It will examine the growth in solar energy production, the photovoltaic effect, operational parameters pertinent to analyzing solar cell performance, categories of solar cells, the Shockley-Queisser limit, and loss mechanisms applicable to current-generation solar cells.

A. Overreliance on Fossil Fuels

In 2019, 80.9% of the world's energy supply originated from fossil fuels like coal, oil, and natural gas, which are non-renewable and polluting [7]. Fossil fuel consumption at this scale results in significant volumes of greenhouse gas emissions, driving global warming and accelerating climate change [4]. Climate change must be addressed as it threatens global food supplies, infrastructure, human habitats, and ecosystems through increasing temperatures, rising sea levels, and severe natural disasters like storms, wildfires, floods, and droughts [8]. Given that global energy demand is expected to increase by 56% by 2040, continued overreliance on fossil fuels will exacerbate the issue of climate change [8].

B. Solar Energy

Adopting renewable energy sources like hydro, wind, and solar can reduce over-dependence on fossil fuels [4]. Until recently, fossil fuels were cheaper to process and transport [9]. Global climate objectives have helped fund research and development, improving generation capacities and increasing integration into existing energy production systems [9]. For example, in 2015, United Nations member states drafted the 2030 Agenda for Sustainable Development Goals, which consists of 17 initiatives to address greenhouse gas emissions [9]. From 2010 to 2021, renewable energy costs decreased, while fossil fuel costs increased [9].

Hydro, wind, and solar are the three most widely utilized renewable energy sources, listed in that order [8]. The renewable energy category also includes geothermal, nuclear, and biomass, but they are difficult to scale [1].

Between 2010 and 2021, solar energy was the fastest-growing renewable in adoption [9]. In the same period, the cost of solar energy declined from \$0.417/kWh to \$0.048/kWh [8]. Moreover, the number of solar installations experienced a 21-fold increase [9]. Solar energy is an appealing energy source because it is abundantly available and globally accessible [4]. Hydro energy is regionally restrictive since it requires elevation changes and high-volume water flow [6]. Wind energy requires high wind speeds [11]. Compared to hydro and wind energy

production, solar causes minimal environmental disturbance [3]. Solar panels and arrays are modular, allowing them to be used for both residential and commercial use. Electrical power generated during the day can be stored in batteries for overnight usage.

C. Solar Cells

1) Photovoltaic Effect

In its simplest form, a solar cell is a single PN junction capable of facilitating the photovoltaic effect [4]. The photovoltaic effect enables the generation of both photovoltage and photocurrent. The photovoltaic effect initiates when sunlight—beams of photons—strikes a solar cell, specifically the PN junction [4]. When the energy of the photons exceeds the band gap energy of the semiconductor, the photons are absorbed within the depletion region [4]. Their energy excites the electrons in the semiconductor, promoting them from the valence band to the conduction band, creating electron-hole pairs [4]. The built-in electric field (arising from the difference in electron concentrations) drives holes and electrons toward the P and N regions, respectively, resulting in photovoltage [4]. As the charge carriers move, they generate photocurrent [4].

2) Operational Parameters

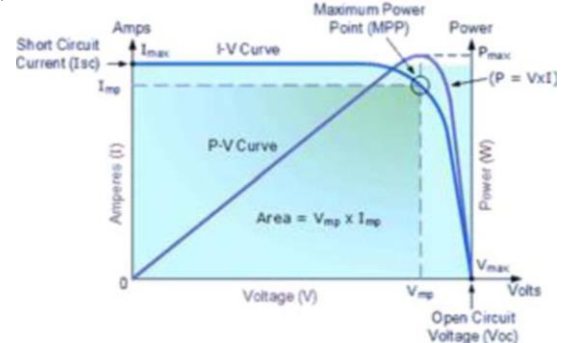


Figure 1. The Current-Voltage (I-V) and Power-Voltage (P-V) Curves of a Typical Solar Cell [4]

The current-voltage (I-V) curve and the power-voltage (P-V) curve of a solar cell are illustrated in Figure 1.

The I-V curve is based on a solar cell's theoretical maximum short-circuit current, I_{SC} , and theoretical maximum open-circuit voltage, V_{OC} , and thus coincides with a solar cell's theoretical maximum power generation capacity, P_{theo} [4].

In contrast, the P-V curve is based on its actual maximum short-circuit current, I_{MP} , and actual maximum open-circuit voltage, V_{MP} , and thus coincides with its actual maximum power generation capacity, P_{max} [4].

In other words, while P_{theo} considers ideal conditions, P_{max} considers real-world conditions. The formulas for P_{theo} and P_{max} are described by (1) and (2), respectively [6].

$$P_{theo} = V_{OC} I_{SC} \quad (1)$$

$$P_{max} = V_{MP} I_{MP} \quad (2)$$

In addition to P_{theo} and P_{max} , the other two most important operational parameters that help understand a solar cell's performance are fill factor, FF and energy conversion efficiency, η [3]. FF is the ratio between the P_{max} and P_{theo} , as seen by (3), whereas η is the ratio between P_{max} and the power of the incident light on the solar cell, P_{in} , as seen by (4) [3].

$$FF = \frac{P_{max}}{P_{theo}} \quad (3)$$

$$\eta = \frac{P_{max}}{P_{in}} \quad (4)$$

Overall, P_{theo} , P_{max} , FF , and η are the most important parameters to consider when analyzing and optimizing a solar cell's performance [3].

Energy conversion efficiency, η , will serve as the basis for examining solar cell performance moving forward in the paper, as optimizing conversion efficiency enables the design of highly scalable and commercially viable solar cell designs [6].

3) Limitations with Current-Generation Solar Cells

Before discussing the limitations of current-generation solar cells and the need for improved solar cell technologies like QDSCs, it is helpful to understand the different types or categories of solar cells.

a) Categories

Solar cells are generally categorized into three generations [4]:

1. First Generation
2. Second Generation
3. Third Generation

First Generation solar cells are the most widely used, owning about 80% share of the solar cell market [4]. They are made from pure crystalline silicon (Si), and although they have relatively low energy conversion efficiencies, which will be discussed soon, they are the most commercially viable [1].

Since pure crystalline Si is expensive to source, Second-Generation solar cells were developed to employ thin films of Si (a few micrometres in thickness) [4]. There are three main types of Second Generation solar cells: amorphous Si, cadmium telluride (CdTe), and copper indium gallium selenide (CIGS) [1]. Although they use less Si and are lighter, they rely on rare materials, are generally less efficient per unit area, and are more expensive to manufacture [1].

QDSCs, with organic, perovskite, and multi-junction solar cells, comprise the Third Generation category [4]. Third Generation solar cells are cutting-edge, research-stage solar cells with the potential to surpass the Shockley-Queisser Limit [1].

First Generation and Second Generation solar cells are considered current-generation solar cells, whereas Third Generation solar cells are referred to as next-generation solar cells.

b) Shockley-Queisser Limit

In 1961, scientists Walter Shockley and Hans Queisser discovered that single-junction solar cells made from crystalline Si (First Generation) have a maximum energy conversion efficiency of approximately 30%, now known as the Shockley-Queisser limit [3]. Their work demonstrated that energy conversion efficiency, η , is also a function of a semiconductor material's bandgap energy, E_{gap} [3]. For instance, the Shockley-Queisser limit is 33.7% for a solar cell made from a semiconductor with a band gap energy of 1.34 eV [4].

It is important to note that the model for the Shockley-Queisser limit has been modified over the years to consider

additional factors [3]. When it was first proposed, it was based on the solar spectrum and the properties of the semiconductor material, e.g., its bandgap energy [3]. Over the years, the model has been improved to consider other variables like semiconductor thickness and Auger recombination processes [3]. Consequently, newer studies often state that the Shockley-Queisser limit of single-junction crystalline Si solar cells is 33% instead of 30% [3]. Additionally, reported conversion efficiencies vary depending on the nature of the study—whether theoretical or experimental—due to oversimplified assumptions or overlooked factors [3].

Other than the notion that solar cells have upper limits on their energy conversion efficiency, the main takeaway from the findings of Shockley and Queisser is that solar cells cannot convert all incident light energy into electrical energy due to different loss mechanisms [3]. A few of the main losses are discussed below.

c) Spectrum Losses

Sunlight consists of photons with different wavelengths, or in other words, different energy levels. Photons with energies below the semiconductor band gap are not absorbed and pass through the semiconductor material. Those with energies above it produce charge carriers with excess kinetic energy, equal to the difference between the energy of the photons and the semiconductor [6]. Due to this kinetic energy, the charge carriers—electrons and holes—have a temperature higher than the semiconductor's crystal lattice [6]. Electrons and holes at these high temperatures are called hot electrons and hot holes [6]. When the hot carriers are absorbed, their excess kinetic energy is lost as heat [6]. In simpler terms, some photons do not have enough energy to generate electricity, while others have too much and lose energy as heat. This mismatch reduces the overall efficiency of a solar cell, as potential incident light energy is lost as heat energy instead of usable electrical energy.

Figure 2 shows how sunlight with certain wavelengths or energies is not absorbed by a typical First Generation solar cell (19%), thermalization losses (33%), and extraction losses (15%), leaving only 33% of the incident light to be used for generating power [12].

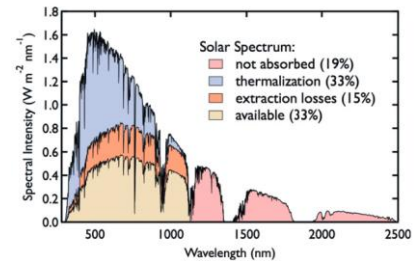


Figure 2. A Spectral Analysis of the Losses Associated with Crystalline Silicon (Si) Solar Cells [12]

d) Charge-Carrier Recombination Losses

Charge carriers must be separate and moving to generate power effectively [1]. As mentioned, electrons and holes move towards different semiconductor regions due to the built-in electric field [4]. Separation is key to creating electric potential or voltage, whereas movement is key to creating current [4]. However, due to different factors like defects on the

semiconductor surface, electrons and holes can recombine, leading to a loss in charge carriers [1]. These recombination events cause a loss in the number of charge carriers and decrease maximum open-circuit voltage and short-circuit current generation potential, decreasing the solar cell's overall energy conversion efficiency.

e) Thermal Radiation Losses

All materials, when heated, emit energy through thermal radiation. As previously described, the solar spectrum consists of photons with varying energy levels. When photons that lack sufficient energy to generate current are absorbed, they contribute to energy loss through thermal radiation. Consequently, some of the absorbed light energy is never converted into electrical energy, reducing the solar cell's overall conversion efficiency.

D. Need for Next-Generation Solar Cells

Solar cells in their current form have limitations that reduce cost-effectiveness. Next-generation solar cell technologies are needed to overcome the typical losses associated with solar cells. Several solar cell technologies with theoretical energy conversion efficiencies far surpassing conventional single-junction solar cells have been proposed, such as the focus of this paper, QDSCs. The next section will discuss quantum dots, the core of QDSC technology. The section after that will discuss QDSCs and how they can change the landscape for solar energy production due to their special characteristics and once enough research is done to resolve their issues.

III. QUANTUM DOTS (QDs)

QDs are extremely small semiconductor crystals with diameters between 1-100 nanometers (nm) [1]. They are bigger than atoms and molecules but significantly smaller than bulk semiconductors like Si [13].

Figure 3 is an image of a QD synthesized using a chemical method called colloidal synthesis. Commonly used QD materials include cadmium sulphide (CdS), lead sulphide (PbS), and gallium arsenide (GaAs) [6]. Bulk semiconductors and QDs can be made from the same material, but their differences come from their size and not the material type.

QDs were discovered by Alexei Ekimov and his peers in 1980 while studying nanocrystals synthesized in glassy matrixes [14]. Their work attracted attention because they noticed that they could change the optical properties of QDs by modifying their size [14]. More importantly, they observed that QDs exhibit the phenomenon of quantum confinement [14]. Ekimov and his peers were awarded the 2023 Nobel Prize in Chemistry for their work [14].

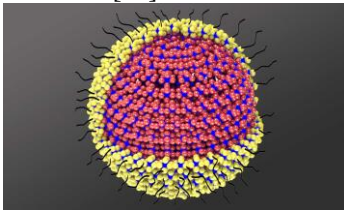


Figure 3. A Quantum Dot (QD) Derived Using Colloidal Synthesis [15]

A. Quantum Confinement

The sizes of QDs are comparable to the Bohr radius of an exciton (a Coulombically bound electron-hole pair) [16]. Due to this nanoscale size, QDs exhibit the phenomenon of quantum confinement, where the motion of charge carriers is restricted in all three spatial dimensions [6]. This confinement results in discrete energy levels, unlike the continuous ones in typical semiconductor materials [6]. Discrete energy levels cause QDs to have remarkable properties, including having a tunable band and being capable of multiple exciton generation (MEG). When discussing QDs, it is imperative to mention quantum confinement.

B. Tunable Bandgap

Due to quantum confinement effects, the band gap energy of quantum dots (QDs) varies inversely with their size [6]. For example, as the size of a QD decreases, its band gap energy increases, enhancing its ability to absorb higher-energy photons. An analogy for this effect is a particle in a box: when the box gets scaled down, the particle bounces around with more energy [14]. As shown in Figure 4, larger QDs have a smaller band gap, while smaller QDs have a larger band gap.

This inverse relationship also influences the optical properties of QDs, as Ekimov and his peers originally observed: larger QDs emit longer wavelengths (red light), whereas smaller QDs emit shorter wavelengths (blue light) [6].

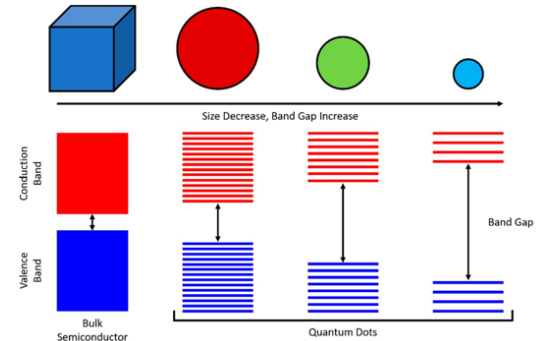


Figure 4. The Effect of Quantum Confinement on the Bandgap of Quantum Dots (QDs) [13]

C. Multiple Exciton Generation (MEG)

Besides a tunable bandgap, the quantum confinement effect enables QDs to generate multiple excitons (electron-hole pairs) from just one high-energy photon [6]. This capability is called multiple exciton generation (MEG) [6].

QDs are capable of MEG because the quantum confinement effect slows the rate at which high-energy charge carriers (hot carriers) lose energy [16]. When photons with energy levels above the bandgap level are absorbed, they generate hot carriers with excess kinetic energy. In bulk semiconductors like Si, these hot carriers quickly relax, releasing energy as heat and resulting in efficiency losses [16]. However, in QDs, hot carriers retain their energy longer due to quantum confinement, permitting them to generate multiple excitons [6], [16]. This process is reflected in Figure 5.

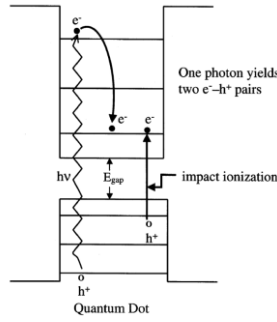


Figure 5. The Effect of Quantum Confinement on Multiple Exciton Generation (MEG) in Quantum Dots (QDs) [16]

The MEG effect occurs easily in QDs but can occur in bulk semiconductors when photons with very high energies are made available, which is impractical [6]. For example, a bulk semiconductor made from lead selenide (PbSe) requires a photon energy threshold of approximately 6.5 times its bandgap energy [17]. In contrast, a QD made from the same materials has a much lower required photon energy threshold of 3.4 times its bandgap energy [17]. The lower threshold in QDs enhances the efficiency of the MEG effect and makes it viable.

IV. QUANTUM DOT SOLAR CELLS (QDSCs)

QDSCs are Third Generation solar cells that take advantage of the unique characteristics of QDs to surpass the Shockley-Queisser limit imposed on their predecessors [4]. QDSCs are said to be solar cells that are sensitized by QDs [17]. QDSCs can capture more of the spectral irradiance radiating from the Sun than conventional solar cells [6]. They achieve this through their ability to be tuned to capture specific wavelengths of light (tunable bandgaps) and generate two or more electron-hole pairs per incoming photon via MEG [6]. QDSCs also have a large extinction coefficient and improved stability against water and oxygen [6].

Despite this, QDSCs remain commercially unviable due to various challenges. The strengths that give QDSCs high efficiency—such as tunable band gaps and MEG—are difficult to realize outside controlled laboratory environments [16]. Although QDSCs can theoretically reach 66% efficiency, according to some studies, similar to multi-junction cells, their current real-world maximum efficiency is around 16.6% [16]. Ongoing research aims to improve this.

A. Structure and Operation

Figure 6 illustrates the structure of a typical QDSC. When photons hit the QDSC and are absorbed by the QDs, electrons are excited, creating electron-hole pairs. QDSCs are especially good at this as QDs can efficiently absorb a wide range of photon energies due to their tunable bandgaps [17]. The electron-hole pairs are rapidly separated at the interface between the QDs and the photoelectrode [17]. The excited electrons move into the photoelectrode material, while the redox couples in the electrolyte cause the holes to be released [17]. The holes in the electrolyte are subsequently transferred to the counter electrode [17]. The flow of electrons toward the photoelectrode and holes to the counter electrode results in the conduction of electricity [17].

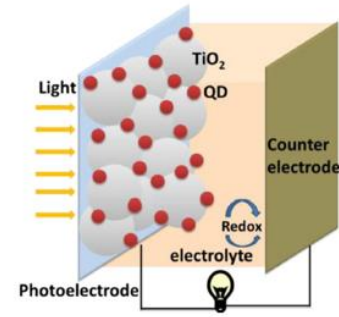


Figure 6. Structure of a Typical Quantum Dot Solar Cell (QDSC) [17]

B. Tunable Bandgap

The quantum confinement effect causes the bandgap energy of QDs to vary inversely with their size, modifying the QDs' absorption spectra [13]. Thus, QDSCs effectively have a tunable bandgap that allows them to capture a wider range of the solar spectrum by only controlling the size of the QDs. QDSCs can capture photons with wavelengths (energies) that cannot be captured with conventional solar cells. A tunable bandgap enables QDSCs to overcome spectrum losses associated with mainstream solar cells.

1) Optimizing Absorption Spectra and Electron Injection Rate

QDs with smaller diameters are sometimes preferred, as they have been shown to support higher electron injection rates than QDs with larger diameters [17]. Smaller QDs increase the available surface area for electron transfer. However, smaller QDs have narrower absorption ranges due to their higher bandgap energies.

Figure 7 illustrates the absorption spectra of QDs synthesized from CdSe, with diameters ranging from 2.3 nm to 3.7 nm.

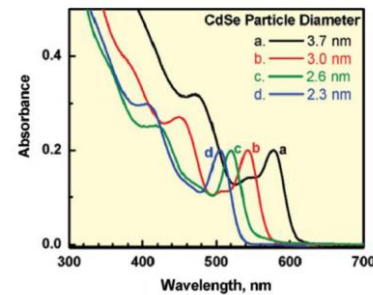


Figure 7. Absorption Spectra of CdSe Quantum Dots (QDs) [17]

As the size of CdSe QDs decreases, their band gap energy increases, allowing them to absorb higher energy (shorter wavelength) photons (see blue curve). Conversely, as the size of the QDs increases, their bandgap energy decreases, allowing them to absorb lower energy (longer wavelength) photons (see black curve). Therefore, although smaller QDs inject electrons into photoelectrodes more efficiently, they miss out on lower-energy photons. Therefore, the trade-off between higher-energy photon absorption via smaller QDs and wider spectral absorption via larger QDs must be carefully considered to optimize energy conversion efficiency.

2) Tunable Multi-Junction Quantum Dot Solar Cells (QDSCs)

Multi-junction solar cells are created by layering or stacking multiple junctions. This approach enables a single cell to capture various portions of the solar spectrum [4].

Multi-junction solar cells can be improved by introducing QDs to create multi-junction QDSCs. QDs can be positioned between layers [18]. The bandgap tunability properties of QDs would extend a multi-junction cell's absorption range, thus increasing the cell's efficiency [12]. Numerical models have shown a potential conversion efficiency of 31.3% for optimized QD multi-junction cells [18]. Correct placement of the QDs is key to ensuring that every layer captures its intended part of the spectrum [18].

C. Multiple Exciton Generation (MEG)

MEG refers to a semiconductor's ability to generate multiple excitons per incident high-energy photon. In conventional solar cells, photons with energies above the bulk semiconductor's bandgap energy are lost as heat, accounting for up to approximately 47% of conversion losses [19]. This significant loss suggests a need for new technologies to capture high-energy photons to contribute to the cell's photocurrent instead of being wasted as heat.

Due to the quantum confinement observed in QDs, energy levels are discretized in QDSCs, such that excess energy is more likely to excite another electron instead of being lost as a phonon, a lattice vibration [20]. This occurs because as QD size decreases, the separation between energy levels increases and surpasses the phonon energy, making it more likely for the incoming photon to generate multiple excitons rather than multiple phonons [20].

As seen in Figure 8, simulations show that when ideal MEG is realized, it can increase energy conversion efficiency by over 10% at a bandgap energy of 0.7 eV. Overall, there is evidence that the QDs in QDSCs can be tuned to have bandgaps for optimal MEG efficiency (0.7-1.3 eV).

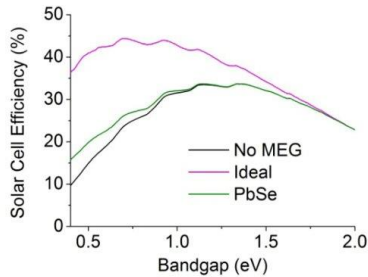


Figure 8. Theoretical QDSC efficiency with and without MEG [20]

Although simulations show great promise for MEG in QDSCs increasing solar cell efficiency, real-world implementations show that MEG has less of an impact on cell efficiency due to fast exciton lifetimes (10-100 picoseconds) [20]. This means that the excitons recombine faster than they can be captured to contribute to the photocurrent. Without fast hole transfer, MEG would have less impact on the efficiency of the cell, leading to high-energy photon energy being lost as heat. Researchers in [21] have demonstrated that fast hole transfer (~4 picoseconds) is possible, allowing extraction of

multiple excitons before recombining—however, more research into strategies to capture these excitons before recombining is needed to improve QDSCs efficiency.

D. Quantum Dot (QD) Layering

Due to the quantum confinement effect, QDs act like wells that confine electrons. However, when closely spaced or layered, electrons can tunnel between them, resulting in improved short-circuit current generation and absorption of a wider range of wavelengths compared to conventional solar cells [22].

The researchers in [22] investigated QD layering using real fabricated solar cells and simulations. They compared ordinary gallium arsenide (GaAs) solar cells to GaAs cells embedded with indium arsenide (InAs) QDs to understand the impact of QD layering on performance. The QD-sensitized cells had varying layers of QDs (up to 40 layers), including some with none.

Table 1. Simulation Results of Quantum Dot (QD) Layering on Short-Circuit Current, Open-Circuit Voltage, Fill Factor, and Energy Conversion Efficiency in Gallium Arsenide (GaAs) Solar Cells [22]

Device	J_{sc} (mA/cm ²)	V_{oc} (volts)	Fill Factor (%)	Efficiency (%)
PN, no QD	22.53	0.983	83.9	14.1
10-layer QDs	25.10	0.993	84.1	15.9
20-layer QDs	26.96	0.996	83.8	17.1
40-layer QDs	29.45	1.001	82.9	18.6

Table 2. Experimental Results of Quantum Dot (QD) Layering on Short-Circuit Current, Open-Circuit Voltage, Fill Factor, and Energy Conversion Efficiency in Gallium Arsenide (GaAs) Solar Cells [22]

Device	J_{sc} (mA/cm ²)	V_{oc} (volts)	Fill Factor (%)	Efficiency (%)
PN, no QD	22.47	1.039	80.0	13.8
10-layer QDs	23.21	0.997	78.5	13.4
20-layer QDs	23.54	0.918	76.8	12.2
40-layer QDs	23.78	0.990	82.3	14.3

External Quantum Efficiency (EQE):
the probability that a photon generates an electron

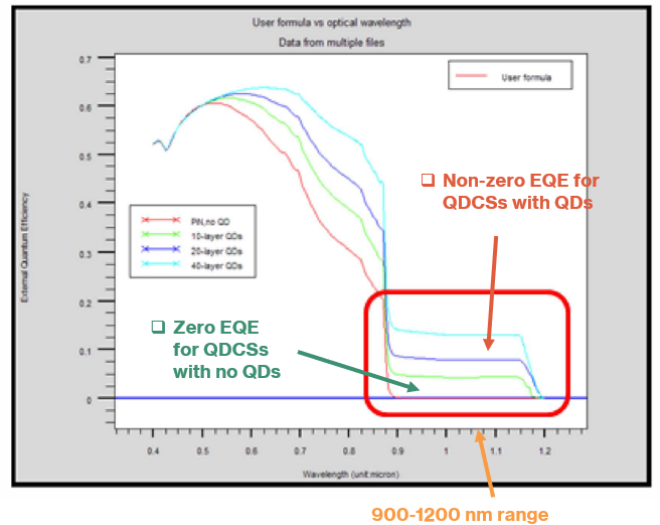


Figure 9. Results of Quantum Dot (QD) Layering on External Quantum Efficiency (EQE) in Gallium Arsenide (GaAs) Solar Cells [22]

Both the simulations and experiments showed similar trends. As seen in Table 1 and Table 2, short-circuit

current increased with the addition of QD layers in simulation and experimental results.

The spectral response was also observed to improve with more QD layers, probably because it allowed the absorption of higher-wavelength (lower-energy) photons. For example, as seen in Figure 9, while the external quantum efficiency (EQE)—the probability that an incident photon generates an electron—is zero for cells without QDs between 900 nm and 1200 nm, it is non-zero for cells with layers of QDs in the same range.

The study also found that controlling QD shapes and sizes during manufacturing is challenging. For example, as seen in Table 2 (experimental results), the 10-layer and 20-layer QDSCs had reduced efficiencies due to manufacturing defects. This trend was not observed in the simulations, as seen in Table 1, as they did not account for the potential for manufacturing defects and other real-world conditions. The defects led to the recombination of charge carriers and reduced open-circuit voltage, which lowers efficiency.

Overall, the study showed that QDs can absorb a broader spectrum of photons and potentially achieve higher efficiency with multiple layers. However, manufacturing challenges must be addressed to realize their full potential.

E. Surface Traps

Surface traps are small imperfections in the structure of the QDs, resulting in reduced solar cell efficiency through non-radiative recombination [2]. At a molecular level, surface traps are atoms on the surface of QDs with dangling or unsatisfied bonds, creating energy levels within the bandgap and capturing free carriers, preventing them from reaching the conduction band and generating current [2]. Efforts must be made to reduce these imperfections that arise during the manufacturing process, and research into improved simulation of surface traps must be conducted to understand the phenomena in more detail.

In traditional QD fabrication methods, strain-induced defects mainly contribute to surface traps caused by lattice mismatches between the QD material and the substrate [2]. Adding additional layers between the QD layers and the substrate has been shown to mitigate this strain by distributing the mechanical stress more evenly [2]. This technique can reduce the amount of surface traps in the material and thus reduce the occurrence of non-radiative recombination, improving the efficiency of the QDSC.

Another method is to add special ligands or coatings to bond with surface atoms [2].

F. Comparison with Alternative State-of-the-Art Solar Cell Technologies

QDSCs and their unique optoelectrical properties suggest a potential to exceed the efficiencies of traditional solar cell technologies. As QDSCs are an emerging PV technology [23], they have not had the time to reach efficiencies of more matured technologies like crystalline Si cells. Figure 10 shows a trend of the best solar cell technology efficiencies since 1975 and highlights how new QDSCs are and their trend for future

efficiency development. From this figure, the best-performing QDSC designs can reach ~19% efficiency.

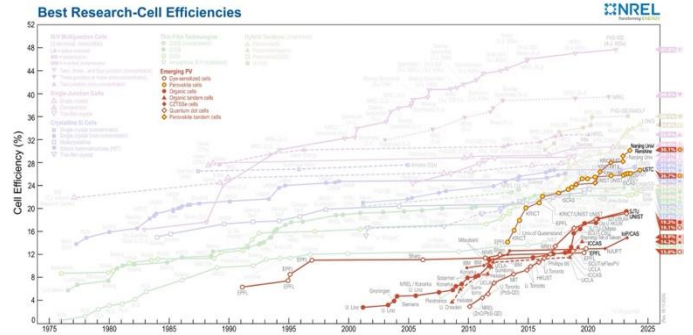


Figure 10. Best research cell efficiencies [23]

Si-based solar cells have dominated the PV market, with monocrystalline single junction silicon cells reaching >26% efficiency in lab tests [24]. This technology has had more opportunity than QDSCs to mature and achieve these impressive efficiencies.

Another emerging PV technology is perovskites. Perovskites are a third-generation PV technology utilizing materials with unique crystal structures, which have been shown to reach efficiencies as high as 26.7% in research environments [23]. Although perovskite solar cells show potential to rival Si-based solar cells, they degrade rapidly when exposed to heat, humidity and solar radiation [25]. These degradation methods are a major problem with perovskite solar cells and are an area where QDSCs perform better. Like QDSCs, the most promising perovskite solar cells utilize lead in their designs to reach high efficiencies [25].

Dye-Sensitized Solar Cells (DSSCs) are another emerging PV technology which has had more time to mature but has only reached efficiencies of ~12% [23]. This technology works similarly to QDSCs, except they use a molecular dye absorber instead of the QD layer and a liquid electrolyte to facilitate charge transport [25]. DSSCs suffer from fast degradation caused by leakage and evaporation of the electrolyte layer and a reduced ability to absorb the solar spectrum compared to QDSCs [25]. Although QDSCs currently have lower efficiencies than perovskite cells, they show potential for improvement with their ability to have tunable bandgap and MEG properties. QDSCs have also shown that they can already surpass the efficiencies of more mature technologies like DSSCs, highlighting the pace of innovation seen with this technology.

G. Environmental and Operational Stability

QDSCs have shown promising efficiencies in lab-based experiments but suffer from various degradation methods that reduce the operational lifetime of the cell [26]. These degradation methods include oxidation, humidity sensitivity, and ultraviolet (UV) light exposure [26]. Tests have indicated that QDSC efficiency can be reduced to 75% of its initial efficiency after only 50 hours of continuous light soaking [26]. Researchers in [27] have shown that these lifetimes could be improved by stabilizing PbS QDs with potassium iodide (KI), maintaining 83% of its original efficiency after 300 hours of

continuous light soaking. The short operational lifetime that current-generation QDSCs face is an area of research that needs more development for the technology to compete with industry-leading silicon PV cells.

H. Manufacturability and Commercial Viability

Numerous manufacturing techniques are used to create QDs and deposit them into QDSCs. In in-situ sensitization, the QDs are sensitized directly on the substrate, resulting in good spatial separation but leaving unreacted products that may affect efficiency [28]. The most common in-situ methods include Successive Ionic Layer Absorption And Reaction (SILAR), hydrothermal synthesis and Chemical Bath Deposition (CBD) [28]. These methods involve introducing the substrate into a solution to deposit the QDs. In-situ methods are ideal for large-scale production due to their one-step fabrication methods [28] but lack control over unreacted products.

Ex-situ (colloidal) sensitization involves creating the QDs in a solution and then depositing that solution onto the substrate, which can reduce defects in the material but is more costly due to the extra steps required [28]. Trade-offs must be made between deposition accuracy, efficiency, and the manufacturability of the QDs to produce QDSCs that can meet the efficiency required for their specific application.

The most promising QDSC designs have utilized Lead (Pb) to achieve high efficiencies, which may curb the technology's ability to become commercially viable [12]. This reliance on an element toxic to humans and the environment is a concern that must be researched further to determine the negative effects of widespread use. In product design, one must consider its full lifecycle from manufacturing to disposal. Proper decommissioning of solar cells is needed to ensure materials can be reused and unsafe elements are contained. Although these considerations must be made, it is important to note that the amount of lead in one lead-acid car battery could be used to make 1000 m^2 of QDSCs to generate 100 kW of power at 10% efficiency with current technologies [12]. Putting the amount of lead needed to make efficient QDSCs using state-of-the-art technologies into perspective, an argument could be made in favour of the commercialization of the technology due to this relatively low requirement for the toxic element.

V. FUTURE WORK

Although QDSCs have shown promising improvements in cell efficiencies in a brief period compared to matured or other emerging PV technologies, several challenges remain to enable QDSCs to reach commercial use. These challenges include improving MEG mechanisms, optimal bandgap tuning, material stability and non-toxic alternatives to QD design [25].

As the technology evolves to solve some of these issues, QDSCs have the potential to become a commercially viable technology capable of reaching efficiencies higher than crystalline silicon. Theoretical MEG calculations show that it could have the ability to improve QDSC efficiency significantly; however, real-world implementation shows challenges in controlling recombination [19]. More research

into MEG optimization advancements is needed to maximize QDSCs performance.

VI. CONCLUSION

QDSCs are a cost-effective next-generation evolution of mainstream crystalline Si solar cells, potentially achieving energy conversion efficiencies double that of their traditional counterparts. The unique properties of QDs, such as quantum confinement, tunable bandgaps, and MEG, enable QDSCs to efficiently absorb a broader spectrum of sunlight and achieve higher electron injection rates. These properties allow QDSCs to overcome fundamental losses associated with traditional solar cells, including the Shockley-Queisser limit.

However, QDSCs face several challenges that must be addressed before commercialization can be fully realized. These include charge-carrier recombination due to surface traps, degradation caused by oxidation and humidity sensitivity, complexities and high manufacturing costs, and the use of toxic materials.

Nevertheless, materials engineering and device optimization advancements have shown promising progress in resolving these issues. Techniques such as QD layering help compensate for recombination events, while surface coatings and ligands can passivate surface traps and improve stability. With these advancements, QDSCs are poised to transition from the research stage to a viable commercial product.

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